

Explaining Frequency Domain CNN Predictions with Grad-CAM: A Comprehensive XAI Approach for Image Classification

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Introduction

- ▶ Traditional CNNs work well with spatial data but have trouble handling high-resolution and large images. Using frequency domain processing with FFT (Fast Fourier Transform) makes CNN training and use faster and more efficient in terms of memory.
- ▶ Explainable AI techniques, like Grad-CAM, are commonly used to understand how models make predictions. They create visual explanations that show which parts of an image are most important for the model's decision.
- ▶ A major research question is whether XAI methods, such as Grad-CAM, can be successfully used with CNNs that operate in the frequency domain?

Literature Review - FFT

Author	Description	Methodology	Results	Limitations
Vasilache et al.	FFT-based CNN convolution implementations using cuFFT and custom fbfft library for GPU performance optimization.	Implemented FFT convolution using cuFFT with autotuning strategy. Developed custom fbfft for batched 1-D/2-D FFTs.	cuFFT: 1.4x-14.5x speedup over cuDNN, up to 23.54x for 13x13 kernels. fbfft: 1.5x over cuFFT, 78% efficiency at 97.5% occupancy.	Poor performance for small 3x3 kernels due to overhead. fbfft limited to square, power-of-2 sizes (8-64 range).
Mathieu, Henaff, LeCun	Fast CNN training with FFTs using Fourier domain convolutions and efficient feature map reuse.	Transform inputs/kernels to frequency domain, perform element-wise multiplications, apply inverse FFT. Custom CUDA implementation with FFT reuse.	2-3x speedup over Torch7/CudaConv. Training: 867ms vs 1756ms. Kernel-size independent with negligible accuracy loss (10^{-5} - 10^{-4}).	Extra GPU memory for frequency representations. Padding for non-power-of-2 inputs. High overhead for small networks.

Literature Review - FDCNN

Author	Description	Methodology	Results	Limitations
Xu, Qin, Sun, Wang, Chen, Ren	DCT-based CNN learning with frequency channel selection for image classification.	RGB→YCbCr→DCT, group into 192 frequency channels, gate module with Gumbel Softmax for dynamic selection.	ImageNet: +1.60% (ResNet-50), +0.63% (MobileNetV2). COCO: +0.8% AP. 87.5% channel pruning with minimal loss.	Fewer channels sometimes better (high-frequency noise). Limited frequency-spatial mapping analysis.
Goh, Chen, Lima	Frequency Domain Convolutional Neural Network (FDCNN) for large image classification using FFT-based convolution operations.	Proposed FDC/FDP layers with RFFT, spatial-to-frequency kernel transform, CIC, artifact removal. Built FDCNN and modified VGG16.	FDCNN: 54.21% faster training, 70.74% memory efficient. Full FDC: 46% faster at 256/512 channels.	Lack of suitable frequency domain activation function limits architecture design. FDCNN shows accuracy decrease when overfitting begins.

Literature Review - XAI

Author	Description	Methodology	Results	Limitations
Ribeiro, Singh, Guestrin	LIME explains black-box predictions by learning local interpretable models. Perturbs input data to understand feature importance.	LIME perturbs inputs locally, fits weighted sparse models to highlight features. SP-LIME uses submodular optimization for dataset explanations.	LIME provides faithful explanations for text/image classifiers. Shows improved trust, model selection, and spurious correlation detection.	Quality depends on model locality. Inadequate for nonlinear regions. High cost for complex models/images.
Scott M. Lundberg, Su-In Lee	SHAP: Shapley values for feature attribution in predictions. Unifies explanation methods under additive framework.	Additive feature attribution as linear sums. Shapley values satisfy accuracy, missingness, consistency axioms. Efficient approximations: Kernel SHAP, Deep SHAP.	SHAP outperforms LIME/DeepLIFT in accuracy and intuition. Kernel SHAP: higher efficiency, faster convergence. Validated on text/image models (MNIST).	Exponential in features; needs approximations. Assumes independence/linearity. High cost for large/correlated inputs.
Iftikhar, Anjum, Siddiqui, Rehman, Ramzan	Interpretable CNN for brain tumor detection using MRIs. Integrates XAI techniques (Grad-CAM, SHAP, LIME) to highlight abnormal tissues and improve transparency.	Two MRI datasets (Msoud: 7023, NeuroMRI: 3264) with preprocessing. CNN iteratively refined using Grad-CAM to remove layers. Explanations via Grad-CAM, SHAP, LIME.	XAI-CNN: 99.21% accuracy/F1 on main dataset, 94.72% on test set. Visualizations show focus on clinically relevant tumor features, outperforming non-XAI approaches.	Limited to MRI; may not generalize to CT/PET. Only tested Grad-CAM, SHAP, LIME. Reduced performance on unseen data due to domain variability.

Literature Review - Grad-CAM

Author	Description	Methodology	Results	Limitations
Selvaraju, Cogswell, Das, Vedantam, Parikh, Batra	Grad-CAM: Visual explanations for CNNs using gradients to create localization maps highlighting important regions for predictions.	Uses gradients from class scores to final conv layer for importance weights. Weighted activation maps with ReLU create localization maps. Combines with Guided Back-propagation for high-res visualizations.	ILSVRC15: 56.51% vs 61.12% error. Human studies: 61.23% vs 44.44%. 0.254 correlation with occlusion. Applied to classification/captioning/VQA. Identifies biases/failures.	Coarse resolution (14×14 for VGG/Alexnet). Needs other methods for high-res. ReLU misses negatives. Limited to decisions.
Chattopadhyay, Sarkar, Howlader, Balasubramanian	Grad-CAM++: Improved visual explanations for CNNs with better object localization and multiple instance handling via pixel-wise gradient weighting.	Weighted positive derivatives of final conv layer w.r.t. class scores. Pixel-wise weights via 2nd/3rd derivatives. Exponential/softmax for smooth differentiation. Combines with Guided Back-prop for high-res.	Outperformed Grad-CAM: 36.84% vs 46.56% drop, 70.72% win rate. Human preference: 109.69 vs 56.08. IoU: 0.38 vs 0.28. Applied to captioning/3D/distillation (+35.5% mAP). Works on VGG-16/AlexNet/ResNet-50.	Higher-order derivatives increase complexity. Variable across architectures/datasets. Coarse maps need upsampling. Limited to positive gradients. Higher overhead than Grad-CAM.

Literature Review - Grad-CAM

Author	Description	Methodology	Results	Limitations
Moujahid, Cherradi, AlSarem, Bahatti, Eljialy, Alsaeedi, Saeed	CNN and Grad-CAM for COVID-19 prediction: Visual explanation framework for automatic classification and interpretation of chest X-ray images (COVID-19, pneumonia, normal).	Merged public X-ray images into balanced 3-class dataset. Used pre-trained VGG16/VGG19/M-NetV2 with transfer learning, added dense/dropout layers. Grad-CAM highlights key regions for prediction interpretation and debugging.	VGG19 best: 96.97% accuracy, 100% precision/F1, 99% recall. Visual explanations matched medically relevant lung areas. No augmentation, validated sources. Outperformed/matched cited methods.	Limited COVID-19 dataset size, possible class imbalance. Needs more samples and hyperparameter tuning. Some models (MobileNetV2) showed overfitting and unreliable Grad-CAM focus.

Datasets Identified

- ▶ *Learning in the Frequency Domain: ImageNet 2012 Large-Scale Visual Recognition Challenge (ILSVRC-2012) dataset, COCO dataset*
- ▶ *Frequency Domain Convolutional Neural Network: Accelerated CNN for Large Diabetic Retinopathy Image Classification: APTOS 2019 blindness detection competition dataset*

Datasets Identified

- ▶ *Visual Explanations from Deep Networks via Gradient-based Localization: ImageNet localization challenge* (ILSVRC-15) dataset, PASCAL VOC 2007 dataset, ImageNet dataset, VQA dataset
- ▶ *Grad-CAM++: Improved Visual Explanations for Deep Convolutional Networks: ImageNet (ILSVRC2012) dataset, PASCAL VOC 2007 dataset, PASCAL VOC 2012 dataset, CIFAR-10 dataset, Flickr30k dataset, Sports-1M dataset*
- ▶ *Explainable CNN for brain tumor detection and classification through XAI based key features identification: Msoud dataset, NeuroMRI dataset*

Research Gaps Identified

- ▶ **Absence of Grad-CAM (or any XAI) for CNNs Trained on Frequency Domain Inputs:** None explained how to interpret or explain the predictions made by these frequency-domain-trained CNNs using XAI methods like Grad-CAM.
- ▶ **Lack of Mapping Explanations from Frequency Domain Features to Original Spatial Images:** No existing research has looked into using Grad-CAM on frequency-domain feature maps from CNNs that were trained using frequency-domain data. The main issue is figuring out how to convert these frequency-based explanations back into the original spatial images to find which pixels contributed to the model's decisions.

Research Gaps Identified

- ▶ **Unexplored Interpretability Challenges for Frequency-Domain Architectures:** The interpretability of models that operate primarily or entirely in the frequency domain is not addressed. How do specific frequency components, identified as "important" by an XAI method, translate to human-understandable visual features in the original image?

Problem Statement

How can the predictions made by a CNN that was trained and tested using images in the frequency domain be clearly explained in terms of the specific areas in the original spatial images, using Grad-CAM and explainable AI techniques?

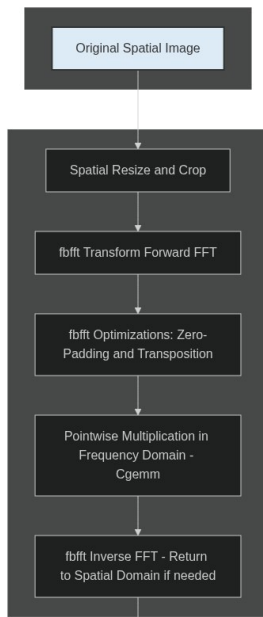
Objectives

1. Adapting Grad-CAM for Convolutional Neural Networks (CNNs) Operating in the Frequency Domain.
2. Creating a strong approach to connect frequency-domain Grad-CAM explanations back to the original image in the spatial domain.
3. Evaluating how clear and accurate the generated spatial explanations are?, both in terms of the amount of information provided and the quality of that information.
4. Studying how specific aspects in the frequency domain affect the ability to understand spatial information?

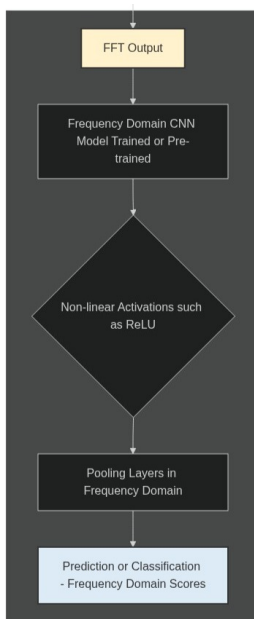
Proposed Methodology

1. Establishing the Frequency-Domain CNN pipeline
 - 1.1 Image data pre-processing for Frequency Domain Input.
 - 1.2 Designing a CNN Architecture for Frequency Domain Inputs.
2. Adapting Grad-CAM for Frequency-Domain CNNs
 - 2.1 Selecting the Target Convolutional Layer.
 - 2.2 Computing Class-Specific Gradients in the Frequency Domain.
 - 2.3 Computing Neuron Importance Weights in the Frequency Domain.
 - 2.4 Generating Raw Frequency-Domain Grad-CAM Map.
3. Mapping Explanations to the Spatial Domain
 - 3.1 Inverse Transform the Frequency-Domain Grad-CAM Map.
 - 3.2 Upsample and Refine the Spatial Explanation Map.
4. Evaluation

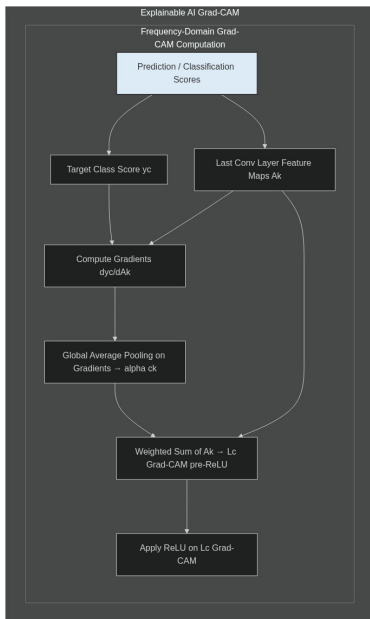
Proposed Architecture: Fast Fourier Transform



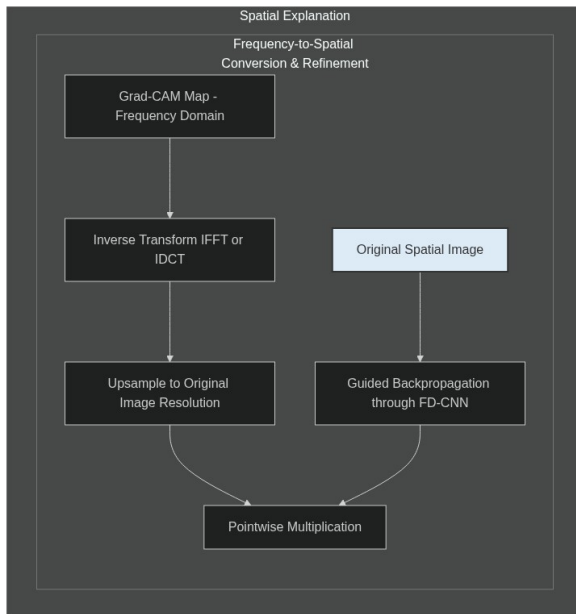
Proposed Architecture: Frequency Domain CNN



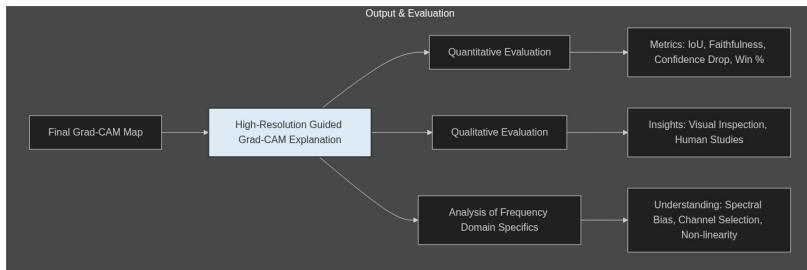
Proposed Architecture: Grad-CAM XAI



Proposed Architecture: Spatial Explanation Generation



Proposed Architecture: Output and Evaluation

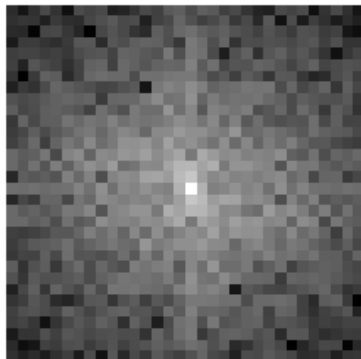


Implementation Results of FFT

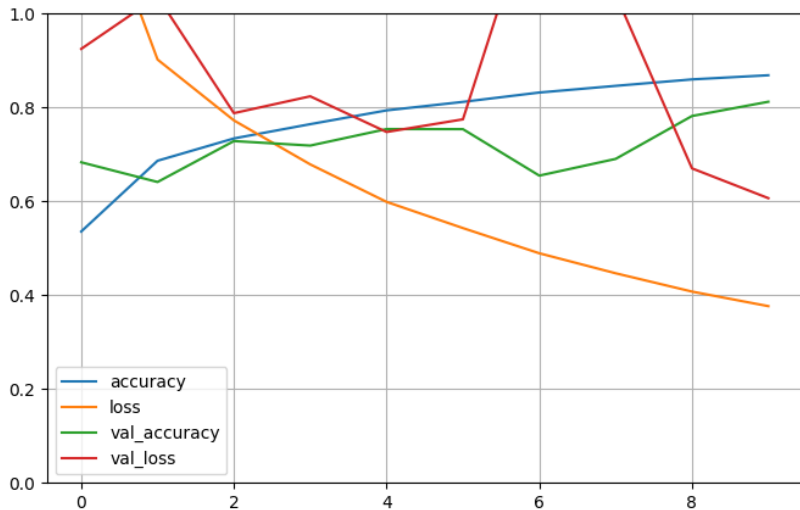
Original Image



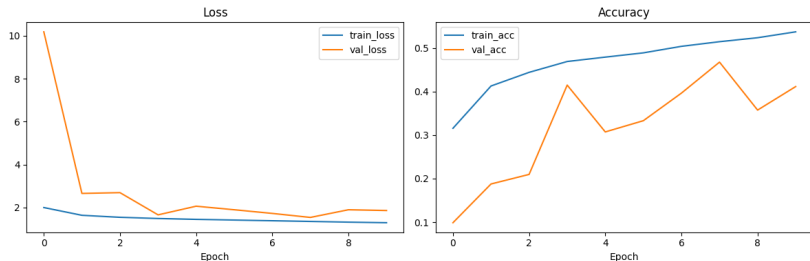
FFT Magnitude Spectrum



Implementation Results of Spatial Domain CNN



Implementation Results of Frequency Domain CNN



Implementation Results of Grad-CAM

Input Image



CAM Image



Conclusion

This project effectively connects the efficiency of frequency domain CNN classification with transparent AI using Grad-CAM. It shows that explanations from XAI can accurately reflect specific areas of the original image, even when models are trained on frequency data. This work improves both the speed and reliability of deep learning systems.

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